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Publications

- [1] **Toward Guarantees for Clinical Reasoning in Vision Language Models via Formal Verification**
Under Review MICCAI 2026
 - **Vikash Singh**, Debargha Ganguly, Haotian Yu, Chengwei Zhou, Prerna Singh, Brandon Lee, Vipin Chaudhary, Gourav Datta
 - [Preprint]
- [2] **CausalGuard: Conformal Inference under Graph Uncertainty**
Under Review NeurIPS 2026
 - **Vikash Singh**, Weicong Chen, Debargha Ganguly, Yanyan Zhang, Nengbo Wang, Sreehari Sankar, Mohsen Hariri, Alexander Nemecek, Chaoda Song, Shouren Wang, Biyao Zhang, Van Yang, Erman Ayday, Jing Ma, Vipin Chaudhary
 - [Preprint]
- [3] **Reliability-Gated Source Anchoring for Continual Test-Time Adaptation**
Preprint, 2026
 - **Vikash Singh**, Debargha Ganguly, Weicong Chen, Sabyasachi Sahoo, Sreehari Sankar, Biyao Zhang, Mohsen Hariri, Shouren Wang, Osama Zafar, Christian Gagné, Vipin Chaudhary
 - [Preprint]
- [4] **VERGE: Formal Refinement and Guidance Engine for Verifiable LLM Reasoning**
ACL 2026
 - **Vikash Singh**, Darion Cassel, Nathaniel Weir, N. Feng, Sam Bayless
 - [Preprint]
- [5] **Trust The Typical**
ICLR 2026
 - Debargha Ganguly, Sreehari Sankar, Biyao Zhang, **Vikash Singh**, Kanan Gupta, Harshini Kavuru, Alan Luo, Weicong Chen, Warren Richard Morningstar, Raghu Machiraju, Vipin Chaudhary
 - [OpenReview]
- [6] **HugRAG: Hierarchical Causal Knowledge Graph Design for RAG**
ICML 2026
 - Nengbo Wang, Tuo Liang, **Vikash Singh**, Chaoda Song, Van Yang, Yu Yin, Jing Ma, Jagdip Singh, Vipin Chaudhary
 - [Preprint]
- [7] **Grammars of Formal Uncertainty: When to Trust LLMs in Automated Reasoning Tasks**
NeurIPS 2025
 - D. Ganguly, **V. Singh**, S. Sankar, B. Zhang, X. Zhang, S. Iyengar, X. Han, A. Sharma, S. Kalyanaraman, V. Chaudhary
 - [Preprint] [Code]
- [8] **Mid-Think: Training-Free Intermediate-Budget Reasoning via Token-Level Triggers**
ACL findings 2026
 - Wang Yang, Debargha Ganguly, Xinpeng Li, Chaoda Song, Shouren Wang, **Vikash Singh**, Vipin Chaudhary, Xiaotian Han
 - [Preprint]
- [9] **Privacy Policy Enforcement Guardrails for Data-Sensitive Retrieval-Augmented Generation**
Preprint, 2026
 - Osama Zafar, Alexander Nemecek, Yiqian Zhang, Wenbiao Li, Debargha Ganguly, **Vikash Singh**, Vipin Chaudhary, Erman Ayday
 - [Preprint]
- [10] **Path-Lock Expert: Separating Reasoning Mode in Hybrid Thinking via Architecture-Level Separation**
Preprint, 2026

- Shouren Wang, Wang Yang, Chuang Ma, Debargha Ganguly, **Vikash Singh**, Chaoda Song, Xinpeng Li, Xianxuan Long, Vipin Chaudhary, Xiaotian Han
 - [Preprint]
- [11] **Overcoming Dynamics-Blindness: Training-Free Pace-and-Path Correction for VLA Models**
Preprint, 2026
- Y. Zhang, C. Song, **V. Singh**, X. Li, K. Ye, Z. Hu, Z. Pu, Y. Yin, V. Chaudhary
 - [Preprint]
- [12] **A Survey on Agent Skills for LLMs: A Lifecycle Perspective from Construction to Ecosystems**
Preprint, 2026
- Wang Yang, Chaoda Song, Xinpeng Li, Shouren Wang, Nengbo Wang, Yanyan Zhang, Chuang Ma, Debargha Ganguly, **Vikash Singh**, Shuai Xu, Jing Ma, Yu Yin, Vipin Chaudhary, Xiaotian Han
 - [Preprint]
- [13] **Efficient Fine-Grained GPU Performance Modeling for Distributed Deep Learning of LLM**
The 32nd IEEE International Conference on High Performance Computing (HiPC), 2025
- Biyao Zhang, Mingkai Zheng, Debargha Ganguly, Xuecen Zhang, **Vikash Singh**, Vipin Chaudhary, Zhao Zhang
 - [Preprint] [Code]
- [14] **K^4 : Online Log Anomaly Detection Via Unsupervised Typicality Learning**
The 32nd IEEE International Conference on High Performance Computing (HiPC), 2025
- W. Chen, **V. Singh**, Z. Rahmani, D. Ganguly, M. Hariri, V. Chaudhary
 - [Preprint] [Code]

Experience

Amazon Web Services

May 2026 – Present

Applied Scientist Intern (Returning)

New York City, NY

- Details will be updated after the internship concludes.
- Manager: **Ali Torakmani**, *Sr. Applied Scientist*, AWS

Amazon Web Services

Aug 2025 – Nov 2025

Applied Scientist Intern

New York City, NY

- Improved logical reasoning of LLMs using formal verification methods, including automated reasoning checks.
- Enhanced the performance of several large language models by over 40% through agentic reasoning frameworks.
- Manager: **Darion Cassel**, *Sr. Applied Scientist*, AWS

MGenio

Jun 2024 – Aug 2024

Machine Learning Internship

Cleveland, OH

- Led self-driven research on machine learning models and their integration on IoT platforms.
- Developed an efficient platform to manage data flow and monitor machine learning model training.
- Designed a pipeline flow and automated data preprocessing system for machine learning models to feed directly into IoT Systems.
- Manager: **Satish Ramade**, *CEO*, MGenio

DRDO, Ministry of Defence

Jan 2022 – Sep 2022

ML/DL Internship

Chandigarh, India

- Developed a sophisticated approach to enhance precision in satellite imagery analysis by employing segmentation, labeling, and training methods using Variational Autoencoders (VAEs), resulting in 83% accuracy.
- Advisor: **Dr. MK Kalra**, *Scientist G, Defence Geoinformatics Research Establishment (DGRE)*, DRDO

Hatchmarine Consultants

Dec 2021 – Feb 2022

Research Intern

Delhi, India

- Developed machine learning models to predict river scour depth in Taiwan, informing strategic resource allocation.
- Fine-tuned predictive models using Python (Scikit-Learn) to achieve high accuracy and meet project requirements.
- Advisor: **Dr. Karan Gupta**, *Technical Director and Founder*

Case Western Reserve University

Jan 2024 – Present

Teaching Assistant (Computational Perception)

Cleveland, OH

- Efficiently grade assignments, ensuring accuracy and providing constructive feedback.
- Deliver engaging lectures on specialized topics, fostering student understanding and conduct effective office hours.

- Professor: **Dr. Michael Lewicki**, *Professor, Dept. of Computer Science & Engineering*, CWRU

Indian Institute of Technology Mandi

Feb 2021 – Aug 2022

Teaching Assistant (Data Science I, II, & III)

HP, India

- Conducted engaging lectures and facilitated Python hands-on lab sessions, enhancing students' practical skills.
- Assessed student understanding through various evaluation methods, including assignment grading and in-person viva sessions.
- Professors: **Dr. Deelip AD, Dr. Varun Dutt, Dr. Manoj Thakur**, *Professors*, IIT Mandi

Education

Case Western Reserve University

Aug 2024 – Present

Doctor of Philosophy in Computer Science

Cleveland, OH

Case Western Reserve University

Aug 2023 – May 2025

Masters of Science in Computer Science (Specialisation in ML/AI)

Cleveland, OH

Indian Institute of Technology Mandi

Jun 2019 – May 2023

Bachelors of Technology in Civil Engineering with minor in AI & Computer Science

Mandi, India

Research Work

Case Western Reserve University

Jan 2024 – Jul 2024

Advancements in XAI with Specialization in Counterfactual Explanation Methods

Cleveland, OH

- Engaged in leading-edge research on Explainable AI, particularly specializing in Counterfactual Explanation methods.
- Advisor: **Dr. Jing Ma**, *Assistant Professor, Department of Computer Science & Engineering*, CWRU

IIT Mandi

Jun 2020 – Jun 2021

Analysis of Nano Particles in Environment using Deep Learning

Mandi, India

- Developed a deep multi-modal architecture for accurately predicting the behavior of nanoparticles on different species using environmental data.
- Advisor: **Dr. Tanushree Parsai**, *Assistant Professor*, IIT Madras

Projects

Enhanced YOLOv4 using SMM on OneAPI in SYCL | *Python3, SYCL, CNN, PyTorch*

Nov 2023

- Developed and integrated Enhanced YOLOv4 with SYCL-Python for advanced object detection algorithms.
- Conducted performance optimization and applied ML techniques to improve real-time data processing.

Human Activity Detector | *Machine Learning, Python3*

Nov 2023

- Built models using Logistic Regression, Decision Tree, and Support Vector Classifier, achieving 96% accuracy with Logistic Regression using accelerometer and gyroscope sensor data.

Landslide Warning System | *Python3, Machine Learning, DNNs*

Aug 2020

- Designed a data-driven predictive system analyzing hillside landslide risk factors like weather, slope, and temperature.

Technical Skills

Languages: Python3, C++, Java, JavaScript

Developer Tools: VS Code, Google Colab, Overleaf, High-Performance Cloud, OneAPI DevCloud

Technologies/Frameworks: TensorFlow, PyTorch, Scikit-Learn, OpenCV, Linux, GitHub

Relevant Coursework

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|---------------------------|--------------------------|---------------------------|-----------------------|
| • Large Language Models | • AI | • Machine Learning | • Pattern Recognition |
| • Data Structures & Algo. | • Analysis of Algorithms | • Data Science I, II, III | • Data Privacy |
| • High-Perf. Systems for | • Deep Learning | • Computer Vision | • Computer Security |

Academic Achievements & Recognitions

- Awarded the **Silver Medal** and **Director's Medal** for academic excellence at IIT Mandi.
- Conducted lab sessions for the "*Training Program on Machine Learning for Ocean Acoustics*" at DRDO-NPOL, Kochi.
- Won two **gold medals** in badminton at the Inter IIT Sports Meet (2019, 2022).
- Secured first place in the **Inter IIT Tech Meet** hackathon at IIT Delhi for plant disease detection algorithm.